A. LONGITUDINAL SECTION — DENSE 16-TAG DUAL-TIER SPATIAL ARRAY IN STEEL CONTAINER

5 mm S355 STEEL CONTAINER WALLS (Acts as Shielded Microwave Cavity — Zero Radiation Leakage)

↑ Upper Wear Rail (8 Tags @ 1.8m) ↑

[A1–A2] Laird Patch Antennas (8.5 dBic)

PRESS CHAMBER

[R1] IMPINJ SPEEDWAY R420

Fixed 4-Port Reader (−86 dBm)

Range: 8–12m Air · 4.0–6.5m Waste

T1 (ID)

T8 (Air)

KNOWN COMPACTED VOLUME V_SMART (±4–5%)

16-Tag Tomographic Grid Resolves Bulk Density & Water Distribution

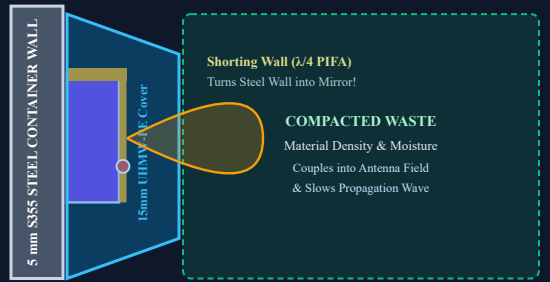
Wear Rail (8 Tags @ 0.9m — Detects Moisture or Water Pooling)

Signal Analysis: Wave Delay indicates Density · Signal Absorption indicates Water Content

(Protected Behind 20mm UHMW-PE Radome)

System Output: Total Payload Weight (kg) + Dry Recyclable Fibre Mass (kg) + Moisture Content (%)

B. DETAIL: HIGH-GRADE ON-METAL PIFA TAG MOUNTING



• 45° Chamfered Wear Rail: Waste Slides Smoothly

• Omni-ID Exo 800 IP68 Tag: +3 dBi Gain ON METAL [10]

• Reading Range: 6–8m on Metal · 15-Year Lifespan